Week 3–5 References

Domain 2: Data Ingestion & Loading (21%)

Databricks Certified Data Engineer Associate

Verified: August 2026

Topics: COPY INTO · Auto Loader · Lakeflow Connect
JDBC/ODBC/REST · Semi-Structured Data · File Formats

Quick Navigation

1. COPY INTO Command
2. Auto Loader Deep Dive
3. Auto Loader Schema Evolution
4. Lakeflow Connect
5. JDBC/ODBC/REST Ingestion
6. Semi-Structured & Nested Data
7. File Formats & Options
8. Ingestion Patterns & Best Practices
9. Release Notes & Updates
10. Decision Framework

COPY INTO Command

Topic	URL
COPY INTO Overview	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-copy-into.html
COPY INTO from Cloud Storage	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
Incremental Loading	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
File Format Options	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-copy-into.html
PATTERN Matching	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
ON_ERROR Handling	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
Idempotent Writes	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html

Auto Loader Deep Dive

Topic	URL
Auto Loader Overview	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
Architecture & How It Works	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
Directory Listing Mode	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
File Notification Mode	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
cloudFiles Configuration	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
cloudFiles.schemaLocation	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
cloudFiles.inferColumnTypes	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
Type Widening (NEW Runtime 16.4)	https://docs.databricks.com/en/data-engineering/schema-evolution.html
cleanSource Option (NEW)	https://docs.databricks.com/en/release-notes/runtime/16.4lts.html
Checkpointing	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
Monitoring & Troubleshooting	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html

Auto Loader Schema Evolution

Topic	URL
Schema Inference	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
Rescue Column	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
schemaEvolutionMode Options	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
addNewColumns	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
addNewColumnsWithTypeWidening	https://docs.databricks.com/en/data-engineering/schema-evolution.html
failOnNewColumns	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
Schema Enforcement vs Evolution	https://docs.databricks.com/en/delta/schema.html
mergeSchema Option	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html
Schema Versioning	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/schema.html

Lakeflow Connect

Topic	URL
Lakeflow Connect Overview	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Standard Connectors	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Managed Connectors	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Query-Based Connectors (GA May 2026)	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Create Connection & Pipeline	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Full Sync & Incremental Sync	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Salesforce Connector	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
ServiceNow Connector	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Database Connectors (SQL, Oracle, PG)	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
UC Governance Integration	https://docs.databricks.com/en/ingestion/lakeflow-connect.html

JDBC/ODBC/REST Ingestion

Topic	URL
JDBC Connection in Notebooks	https://docs.databricks.com/en/ingestion/index.html
JDBC to Delta Table	https://docs.databricks.com/en/ingestion/index.html
JDBC Driver Setup	https://docs.databricks.com/en/ingestion/index.html
Connection Pooling	https://docs.databricks.com/en/ingestion/index.html
ODBC with Databricks	https://docs.databricks.com/en/ingestion/index.html
Configure ODBC Connection	https://docs.databricks.com/en/ingestion/index.html
REST API Calls in Notebooks	https://docs.databricks.com/en/ingestion/index.html
JSON from REST Endpoints	https://docs.databricks.com/en/ingestion/index.html
Pagination Patterns	https://docs.databricks.com/en/ingestion/index.html
Error Handling & Retries	https://docs.databricks.com/en/ingestion/index.html

Semi-Structured & Nested Data

Topic	URL
Reading JSON Files	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/patterns.html
JSON with Auto Loader	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/patterns.html
Nested JSON	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/patterns.html
Struct Data Type	https://docs.databricks.com/en/sql/language-manual/sql-ref-data-types-struct.html
Array Data Type	https://docs.databricks.com/en/sql/language-manual/sql-ref-data-types-array.html
EXPLODE Arrays	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
VARIANT Data Type (Semi-Structured)	https://docs.databricks.com/en/sql/language-manual/sql-ref-data-types-variant.html
Reading XML Files	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/patterns.html
CSV with Embedded JSON	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/patterns.html

File Formats & Options

Format	URL
CSV Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-csv.html
JSON Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-json.html
Parquet Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-parquet.html
Avro Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-avro.html
ORC Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-orc.html
Text Format	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-text.html
Parquet Compression	https://docs.databricks.com/en/sql/language-manual/sql-ref-format-parquet.html

Ingestion Patterns & Best Practices

Pattern	URL
Batch vs Streaming Ingestion	https://docs.databricks.com/en/ingestion/index.html
Incremental Load Concepts	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
CDC (Change Data Capture)	https://docs.databricks.com/en/delta/change-data-feed.html
Watermark-Based Ingestion	https://docs.databricks.com/en/structured-streaming/index.html
Idempotent Pattern	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html
Error Handling & Validation	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-copy-into.html
Quarantine Pattern	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
Partition Strategy for Ingestion	https://docs.databricks.com/en/ingestion/cloud-object-storage/auto-loader/index.html
Cost Optimization	https://docs.databricks.com/en/ingestion/cloud-object-storage/copy-into.html

Release Notes & Updates

Feature/Version	URL
Databricks Runtime 16.4 LTS	https://docs.databricks.com/en/release-notes/runtime/16.4lts.html
Type Widening (Runtime 16.4)	https://docs.databricks.com/en/release-notes/runtime/16.4lts.html
cleanSource (Runtime 16.4)	https://docs.databricks.com/en/release-notes/runtime/16.4lts.html
Query-Based Connectors GA (May 2026)	https://docs.databricks.com/en/ingestion/lakeflow-connect.html
Databricks Runtime 17.0	https://docs.databricks.com/en/release-notes/runtime/17.0.html

Summary: 65+ official Databricks documentation URLs covering Domain 2 (21% of exam). Verified August 2026 against May 2026 Exam Guide.